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Forecasting in Data-Rich Environment

Does the S&P 500–Treasury Interaction Improve U.S. Industrial Production Forecasts?

Evidence from a Data-Rich Elastic Net Framework

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1 Introduction

Financial markets can provide information about expected economic conditions. Stock prices react to expectations about future cash flows, business conditions, and risk. Interest rates affect discount rates and financing conditions. Fama (1990) finds that stock returns are related to future production growth, with a stronger relationship at longer horizons.

Chen et al. (1986) show that industrial production and interest-rate spreads are among the macroeconomic variables related to stock returns. Ratanapakorn and Sharma (2007) find a negative relationship between U.S. stock prices and long-term interest rates. These findings suggest that stock market and interest-rate information may be useful when used together.

A relationship between financial variables and economic activity does not necessarily mean that these variables improve out-of-sample forecasts. Fama (1990), for example, notes that strong in-sample relationships may become weaker when new data are used. However, Fama (1990) does not test whether an explicit interaction between S&P 500 returns and changes in the 10-year Treasury yield improves out-of-sample forecasts of industrial production growth.

The main aim of this report is to test whether stock returns, Treasury movements, and their interaction provide useful out-of-sample forecasting information. The study examines the predictive value of the S&P 500–Treasury interaction and tests whether it provides additional forecasting information beyond the individual financial variables. The models also include a broad set of macroeconomic variables from the Federal Reserve Economic Data–Monthly Database (FRED-MD).

An AR(1) model of industrial production growth is used as the benchmark. The competing models combine the macroeconomic data with daily S&P 500 returns, daily Treasury-yield changes, and their interaction. The models are estimated using Elastic Net regularization because FRED-MD contains many correlated predictors (Zou and Hastie, 2005).

The analysis evaluates point forecasts of industrial production growth using the mean squared forecast error (MSFE) and Theil's U. The Diebold–Mariano test and the Clark–West test are used to assess the statistical significance of differences in forecast performance. Because conditional-mean forecasts may not capture tail and upside dynamics of industrial production growth, this study also considers quantile forecasts to examine whether the financial variables provide different information across the distribution of future growth, especially during periods of weak industrial production growth.

The rest of the paper is organized as follows. Section 2 describes the data and variable construction. Section 3 explains the forecasting models and evaluation methods. Section 4 presents the point-forecast results. Section 5 presents the quantile forecasting analysis. The final section summarizes the main findings and concludes.

2 Problem and Data Description

The analysis combines monthly U.S. macroeconomic data with daily financial data from the S&P 500 and the 10-year U.S. Treasury yield. The sample covers January 1990 to June 2026.

2.1 Macroeconomic Data

The macroeconomic information set comes from the Federal Reserve Economic Data–Monthly Database (FRED-MD), a large monthly U.S. macroeconomic dataset associated with the Federal Reserve Bank of St. Louis's FRED data system. The database contains 127 monthly macroeconomic series covering output and income, the labor market, housing, consumption, orders and inventories, money and credit, interest and exchange rates, prices, and the stock market. Using a broad information set allows the models to use information from different parts of the economy rather than a small number of selected predictors.

The recommended FRED-MD transformations are applied before the variables enter the forecasting models (McCracken and Ng, 2016). These transformations reduce problems such as trends and non-stationarity and make the series more suitable for time-series analysis. Appendix A lists the FRED-MD variables used in the study, and Appendix B gives the transformation codes.

2.2 Research Question and Objectives

The main research question of this study is:

- ***Does the interaction between S&P 500 returns and changes in the 10-year U.S. Treasury yield provide additional predictive information for out-of-sample forecasts of U.S. industrial production growth, beyond their individual effects and a broad macroeconomic information set?***

The objective of the study is to assess the predictive value of financial-market information for future U.S. industrial production growth. The specific objectives are:

1. To examine whether S&P 500 returns and changes in the 10-year U.S. Treasury yield provide useful information for forecasting U.S. industrial production growth.
2. To determine whether the interaction between S&P 500 returns and changes in the 10-year U.S. Treasury yield provides additional predictive information beyond the individual financial variables and the broader macroeconomic information set.
3. To compare the out-of-sample forecast accuracy of the competing models across 1-, 3-, 6-, 12-, and 24-month forecast horizons and across alternative rolling-window lengths, using mean squared forecast error, Theil's U, and the Diebold–Mariano test relative to the AR(1) benchmark.
4. To assess the incremental predictive content of the financial variables within the nested Elastic Net specifications using the Clark–West with particular attention to whether the S&P 500–Treasury interaction improves forecasts beyond the model containing their individual effects.
5. As an additional analysis, to examine whether the predictive value of financial information differs across the conditional distribution of future industrial production growth using quantile forecasts.

2.3 Industrial Production Growth

The target variable is monthly U.S. industrial production growth. Industrial production is a common measure of real economic activity.

Let IP_t denote the Industrial Production Index in month t. Monthly industrial production growth is calculated as

$$IPG_t = 100\,[\ln(IP_t) - \ln(IP_{t-1})]\,, \tag{1}$$

where IPG_t is the monthly percentage log growth rate of industrial production.

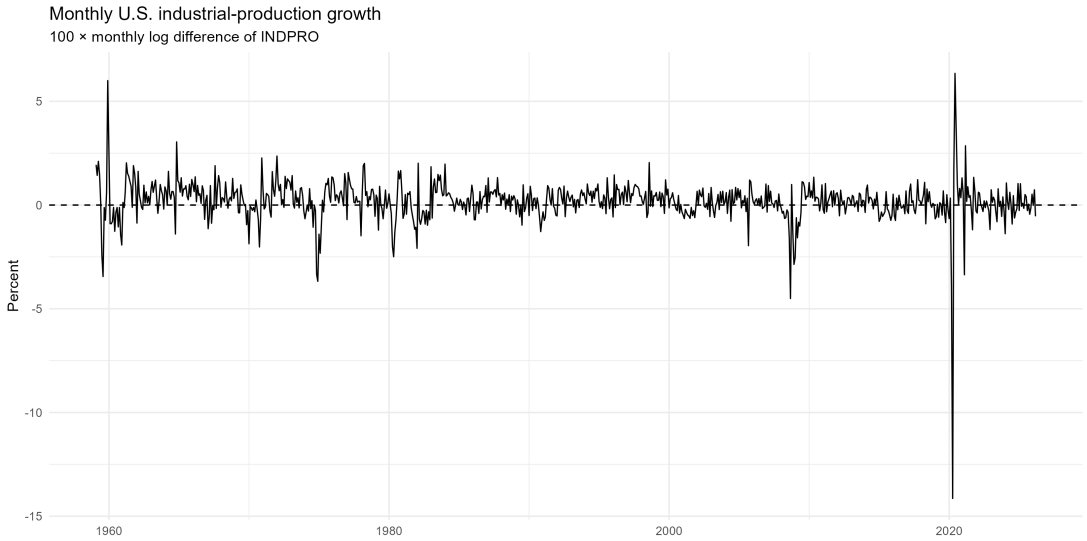


Figure 1: Monthly U.S. Industrial Production Growth

Figure 1 shows monthly industrial production growth over the sample period. Growth is usually close to zero during stable periods. Larger movements occur during periods of economic disruption. The effects of the 2008 financial crisis and the COVID-19 pandemic are clearly visible in the figure. The largest movements occur around 2020, when industrial production fell sharply and then recovered strongly.

2.4 Financial Variables

The financial information is added separately through daily S&P 500 returns, movements in the 10-year U.S. Treasury yield, and an interaction between the two variables. To avoid using the same financial information twice, the corresponding stock-market and interest-rate variables are removed from the FRED-MD predictor set. The remaining variables are referred as **Restricted FRED-MD predictors**.

2.4.1 S&P 500 Return

Daily S&P 500 closing prices are obtained from Yahoo Finance (Yahoo Finance, 2026). The S&P 500 is used as a broad measure of movements in the U.S. equity market.

Let P_d denote the closing value of the S&P 500 on trading day d. The daily percentage log return is calculated as

$$R_d^{SP} = 100\left[\ln(P_d) - \ln(P_{d-1})\right], \tag{2}$$

where R_d^{SP} is the daily percentage log return.

2.4.2 10-Year U.S. Treasury Yield

The second financial variable is the 10-year U.S. Treasury yield. The series is obtained from FRED and is published by the Board of Governors of the Federal Reserve System (Board of Governors of the Federal Reserve System (US), 2026).

Let Y_d^{10} denote the 10-year Treasury yield on trading day d. The daily percentage log change used as the Treasury predictor is calculated as

$$G_d^{10} = 100\left[\ln(Y_d^{10}) - \ln(Y_{d-1}^{10})\right], \tag{3}$$

where G_d^{10} is the daily proportional change in the long-term Treasury yield.

2.4.3 Stock–Treasury Interaction

The Stock–Treasury interaction is used to test whether the joint movement of stock returns and long-term interest rates contains information about future industrial production growth beyond the information in the two variables separately.

The interaction is first constructed at the daily frequency as

$$INT_d = R_d^{SP} \times G_d^{10}, \tag{4}$$

where R_d^{SP} is the daily S&P 500 return defined in Equation (2), and G_d^{10} is the daily proportional change in the long-term Treasury yield defined in Equation (3).

The first difference is used to make the Stock-Treasury interaction term stationary and it is calculated as

$$\Delta INT_d = INT_d - INT_{d-1}, \tag{5}$$

The transformed interaction selected by this procedure is used in the forecasting models.

2.5 Data Transformation and Alignment

Industrial production and Restricted FRED-MD predictors are monthly, while the S&P 500 and 10-year Treasury variables are daily. The daily financial observations are kept separately and are not converted into monthly averages or sums.

For each month, the actual trading days are ordered as $D01, D02, \ldots, D_{\max}$. Here, $D01$ is the first trading day of the month, and $D_{\max}$ is the largest number of trading days observed in any month. Because months have different numbers of trading days, a trading-day position that does

not exist is recorded as zero with an availability indicator of zero. An observed trading day has an availability indicator of one.

Weekends and market holidays are treated as non-trading days, not as missing values. If a value is missing on an actual trading day, the last available value is carried forward using last observation carried forward (LOCF). Future values are not used to fill missing observations.

At the end of month t, where t is the forecast month, all information observed during that month is assumed to be available. The predictor vector $\mathbf{x}_t$ contains the FRED-MD variables and the daily financial information available in month t. It is used to forecast industrial production growth y_{t+h}, where h is the forecast horizon in months.

3 Methods

3.1 Model Specifications

The forecasting models are built by adding financial information to the Restricted FRED-MD predictor set.

- **AR(1)**
- **EN-1:** Restricted FRED-MD predictors.
- **EN-2:** Restricted FRED-MD predictors + daily S&P 500 return.
- **EN-3:** Restricted FRED-MD predictors + daily 10-year Treasury log-change.
- **EN-4:** Restricted FRED-MD predictors + daily S&P 500 return + daily Treasury log-change.
- **EN-5:** Restricted FRED-MD predictors + daily S&P 500 return + daily Treasury log-change + daily Stock–Treasury interaction term.

3.2 Autoregressive Benchmark

AR(1) is the benchmark model for industrial production growth. A separate direct model is estimated for each forecast horizon h. The AR(1) model is

$$y_{t+h} = c_h + \phi_h y_t + \varepsilon_{t+h}, \qquad h \in \{1, 3, 6, 12, 24\}, \tag{6}$$

where y_t is current industrial production growth and y_{t+h} is growth h months ahead (Hyndman and Athanasopoulos, 2021). This benchmark shows whether the larger macroeconomic and financial information sets improve on the recent history of industrial production alone.

3.3 Elastic Net

The Elastic Net (EN), proposed by Zou and Hastie (2005), is a regularized regression method that combines Least Absolute Shrinkage and Selection Operator(LASSO) and ridge regression. It is useful when there are many predictors and some of them are highly correlated. Elastic Net shrinks the estimated coefficients and can set less useful coefficients to zero. Unlike LASSO, it can also

keep groups of correlated predictors (Zou and Hastie, 2005). This is useful in macroeconomic forecasting because many economic variables contain similar information.

A separate direct Elastic Net model is estimated for each forecast horizon. The forecasting model is

$$y_{t+h} = \beta_{0,h} + \mathbf{x}_t'\boldsymbol{\beta}_h + \varepsilon_{t+h}, \qquad h \in \{1, 3, 6, 12, 24\}, \tag{7}$$

where y_{t+h} is U.S. industrial production growth h months ahead, $\mathbf{x}_t$ is the predictor vector available at forecast month t, $\beta_{0,h}$ is the intercept for horizon h, and $\boldsymbol{\beta}_h$ is the vector of coefficients. Depending on the model, the predictor vector contains the Restricted FRED-MD variables, the S&P 500 return, the 10-year U.S. Treasury variable, and their interaction.

For each forecast horizon, the EN estimator is obtained by minimizing

$$(\hat{\beta}_{0,h}, \hat{\boldsymbol{\beta}}_h) = \arg\min_{\beta_{0,h}, \boldsymbol{\beta}_h} \left\{ \frac{1}{2N_h} \sum_{t \in \mathcal{T}_h} \left(y_{t+h} - \beta_{0,h} - \mathbf{x}_t'\boldsymbol{\beta}_h\right)^2 + \lambda_h \left[\alpha_h \sum_{j=1}^{p} |\beta_{h,j}| + \frac{1-\alpha_h}{2} \sum_{j=1}^{p} \beta_{h,j}^2 \right] \right\}, \tag{8}$$

where $\mathcal{T}_h$ is the set of training observations for horizon h, N_h is the number of observations in the training sample, and p is the number of predictors. The first part measures the difference between the observed and fitted values. The second part is the penalty on the regression coefficients.

The parameter $\lambda_h \geq 0$ controls the strength of the penalty for horizon h. A small λ_h gives little shrinkage. As λ_h increases, the coefficients are shrunk more strongly toward zero.

The parameter α_h, where $0 \leq \alpha_h \leq 1$, controls the mix of the two penalties. When $\alpha_h = 1$, EN becomes LASSO and uses only the L_1 penalty, which can set some coefficients exactly to zero. When $\alpha_h = 0$, EN becomes ridge regression and uses only the L_2 penalty, which shrinks coefficients but usually does not remove variables. Values between zero and one combine both penalties (Zou and Hastie, 2005).

After estimation, the direct h-month-ahead forecast is

$$\hat{y}_{t+h|t} = \hat{\beta}_{0,h} + \mathbf{x}_t'\hat{\boldsymbol{\beta}}_h, \qquad h \in \{1, 3, 6, 12, 24\}. \tag{9}$$

The predictors are standardized before estimation because the penalty depends on their scale. The intercept is not penalized. The values of α_h and λ_h are selected using the training data. This controls model complexity and reduces the risk of overfitting when many predictors are available.

Hyperparameter Tuning

Following Zou and Hastie (2005), α is fixed at each value in a grid from 0.1 to 0.9, while λ is selected by cross-validation. For each value of α, the corresponding λ is selected using the training data. The combination with the lowest validation MSFE is retained.

3.4 Forecast Evaluation

Mean Squared Forecast Error

Forecast accuracy is evaluated using the Mean Squared Forecast Error (MSFE), which measures the average squared difference between the observed and forecast values of industrial production growth. For forecast horizon h, MSFE is calculated as

$$\text{MSFE}_h = \frac{1}{N_h} \sum_{t=1}^{N_h} \left(y_{t+h} - \hat{y}_{t+h|t} \right)^2, \tag{10}$$

where y_{t+h} is the observed industrial production growth, $\hat{y}_{t+h|t}$ is its forecast, and N_h is the number of out-of-sample forecasts at horizon h. A lower MSFE indicates better forecast accuracy. In this study, MSFE is used to compare the competing models at the 1-, 3-, 6-, 12-, and 24-month forecast horizons. (Hyndman and Athanasopoulos, 2021; West, 2006).

Theil's U

Theil's U compares the MSFE of each model with the AR(1) benchmark (Theil, 1966):

$$U_h = \sqrt{\frac{\text{MSFE}_h^{\text{Model}}}{\text{MSFE}_h^{\text{AR(1)}}}}. \tag{11}$$

A value below one favours the competing model, while a value above one favours AR(1).

3.5 Forecast Accuracy Tests

Two tests are used to check whether differences in forecast accuracy are statistically significant.

3.5.1 Diebold–Mariano Test

The Diebold–Mariano (DM) test compares each Elastic Net model with AR(1) (Diebold and Mariano, 1995). Under squared-error loss,

$$d_t = e_{AR,t+h}^2 - e_{EN,t+h}^2, \tag{12}$$

with the null hypothesis

$$H_0 : E(d_t) = 0. \tag{13}$$

A positive statistic favours the Elastic Net model, while a negative statistic favours AR(1). For multi-step forecasts, a heteroskedasticity and autocorrelation consistent (HAC) variance estimator with lag $h - 1$ is used. The Harvey–Leybourne–Newbold correction is applied for small samples (Harvey et al., 1997):

$$DM_{\text{HLN}} = DM \left[\frac{N + 1 - 2h + \frac{h(h-1)}{N}}{N} \right]^{1/2}. \tag{14}$$

The DM test compares EN-1 through EN-5 with AR(1) at every forecast horizon for the 120-, 180-, and 240-month rolling windows.

3.5.2 Clark–West MSFE-Adjusted Test

The Clark–West test is used for nested Elastic Net models (Clark and West, 2007). Let S denote the smaller model and L denote the larger nested model. The adjusted loss difference is

$$f_{t+h} = e^2_{S,t+h} - \left[e^2_{L,t+h} - \left(\hat{y}_{S,t+h|t} - \hat{y}_{L,t+h|t}\right)^2\right]. \tag{15}$$

This is equivalent to comparing the MSFE of the smaller model with the adjusted MSFE of the larger model:

$$\overline{f} = \text{MSFE}_S - \left[\text{MSFE}_L - \frac{1}{P}\sum_{t=1}^{P}\left(\hat{y}_{S,t+h|t} - \hat{y}_{L,t+h|t}\right)^2\right]. \tag{16}$$

The test statistic is

$$CW = \frac{\sqrt{P}\,\overline{f}}{\sqrt{\widehat{\text{Var}}_{\text{NW}}(f)}}. \tag{17}$$

The Newey–West calculation includes autocovariances through lag $h-1$. The test is one-sided, and positive values favour the larger model. The 10% and 5% critical values are 1.282 and 1.645, respectively.

The Clark–West test is applied only to valid nested Elastic Net models using the main 180-month rolling window. The EN-4 versus EN-5 comparison directly tests whether the S&P 500–Treasury interaction adds information after the macroeconomic, S&P 500, and Treasury variables are already included.

3.6 Statistical Assumptions

The forecasting framework relies on the following statistical assumptions:

- **Predictive interpretation:** The models are predictive rather than causal. The estimated coefficients are used to generate forecasts and are not interpreted as causal effects of the predictors on industrial production growth.
- **Time-series stability:** The transformed variables are assumed to have sufficiently stable time-series properties for forecasting. The transformations applied to the data are intended to reduce trends and unit-root behavior that could produce misleading forecasting relationships.

4 Empirical Analysis

The empirical analysis starts with point forecasts of industrial production growth. The 180-month rolling window is the main specification. The analysis also uses 120- and 240-month windows to check whether the results depend on the estimation-window length.

4.1 Point-Forecast Results: 180-Month Rolling Window

Table 1 reports the point-forecast results for the main 180-month rolling window.

AR(1) is difficult to outperform. It has the lowest MSFE at the 3-, 6-, 12-, and 24-month horizons. At 12 months, the difference is very small: AR(1) has an MSFE of 2.060, while the Elastic Net models range from 2.069 to 2.080. The main exception is the one-month

Table 1: Point-Forecast Performance: 180-Month Rolling Window

	1 Month		3 Months		6 Months		12 Months		24 Months	
Model	**MSFE**	**U**	**MSFE**	**U**	**MSFE**	**U**	**MSFE**	**U**	**MSFE**	**U**
AR(1)	2.613	1.000	**2.547**	1.000	**2.208**	1.000	**2.060**	1.000	**2.351**	1.000
EN-1	2.993	1.070	4.532	1.334	2.585	1.082	2.069	1.002	3.343	1.192
EN-2	2.948	1.062	3.043	1.093	2.463	1.056	2.070	1.002	4.181	1.334
EN-3	2.811	1.037	4.627	1.348	2.761	1.118	2.073	1.003	3.104	1.149
EN-4	2.761	1.028	3.029	1.091	2.661	1.098	2.079	1.005	3.529	1.225
EN-5	**2.457**	**0.970**	2.741	1.037	2.862	1.138	2.080	1.005	3.638	1.244

Notes: MSFE denotes mean squared forecast error, and U denotes Theil's U relative to AR(1). Lower values indicate better forecast accuracy. A value of $U < 1$ means that the model has a lower MSFE than AR(1).

horizon. EN-5, which includes the Restricted FRED-MD predictors, S&P 500 returns, Treasury information, and their interaction, has an MSFE of 2.457, compared with 2.613 for AR(1). Its Theil's U is 0.970, which also indicates a lower MSFE than the benchmark. This shows a modest short-run numerical gain from the full information set.

AR(1) remains a strong benchmark for conditional-mean forecasts. The next step checks whether this conclusion changes when shorter and longer rolling windows are used.

4.2 Robustness Across Rolling Windows

Table 1 shows that AR(1) has the lowest MSFE at four of the five horizons in the 180-month window. To check whether this result depends on the window length, the analysis is repeated with 120- and 240-month rolling windows. The results are shown in Table 2.

Table 2: Point-Forecast MSFE Across Alternative Rolling Windows

	1 Month		3 Months		6 Months		12 Months		24 Months	
Model	**120**	**240**	**120**	**240**	**120**	**240**	**120**	**240**	**120**	**240**
AR(1)	**4.385**	**2.495**	**3.092**	2.516	3.118	**2.162**	**2.271**	**2.065**	**2.400**	**2.356**
EN-1	4.721	4.921	14.406	3.273	13.533	2.713	6.677	2.068	25.282	2.467
EN-2	6.167	4.888	11.515	2.479	11.275	2.321	7.310	2.069	25.132	2.703
EN-3	7.121	4.076	9.939	3.761	**2.682**	2.518	9.232	2.072	7.015	2.530
EN-4	5.440	4.220	8.575	**2.466**	3.916	2.350	10.426	2.075	6.800	2.650
EN-5	5.164	3.533	9.365	2.511	6.211	2.537	6.090	2.067	5.237	2.703

Notes: Entries report mean squared forecast error (MSFE). Columns 120 and 240 refer to the rolling-window lengths in months. Lower values indicate better forecast accuracy. Bold values identify the lowest MSFE for each horizon and rolling window. Results for the main 180-month window are reported in Table 1. All windows are evaluated over the common out-of-sample period.

The 120-month window gives a similar result. AR(1) has the lowest MSFE at the 1-, 3-, 12-, and 24-month horizons. The only exception is six months, where EN-3 has an MSFE of 2.682, compared with 3.118 for AR(1).

The 240-month window also gives a similar result. AR(1) has the lowest MSFE at the 1-, 6-, 12-, and 24-month horizons. At three months, EN-4 has the lowest MSFE at 2.466, compared with 2.516 for AR(1), although the difference is small.

Across the 120-, 180-, and 240-month windows, AR(1) has the lowest MSFE in 12 of the 15 horizon–window combinations. The three exceptions are EN-3 at six months with the 120-month window, EN-5 at one month with the 180-month window, and EN-4 at three months with the 240-month window. This shows that the main result does not depend on the rolling-window length: **AR(1) remains difficult to outperform** consistently.

MSFE values do not show whether the differences are statistically significant. Therefore, each Elastic Net model is compared with AR(1) using the Diebold–Mariano test for all three rolling windows.

4.3 Predictive Accuracy Relative to AR(1): Diebold–Mariano Tests

The Diebold–Mariano (DM) test compares each Elastic Net model with AR(1) at every horizon and rolling window. Positive statistics favour the Elastic Net model, while negative statistics favour AR(1). The reported p-values are two-sided, and the statistics include the Harvey–Leybourne–Newbold small-sample correction.

Table 3: Diebold–Mariano Tests Against AR(1) Across Rolling Windows

		1 Month		3 Months		6 Months		12 Months		24 Months	
Window	**Model**	**DM**	p	**DM**	p	**DM**	p	**DM**	p	**DM**	p
120	EN-1	-0.374	0.709	-1.025	0.307	-1.021	0.309	-1.215	0.226	-0.966	0.336
	EN-2	-0.880	0.380	-1.019	0.310	-1.040	0.300	-1.278	0.203	-1.047	0.297
	EN-3	-0.870	0.386	-1.069	0.287	0.505	0.614	-1.134	0.258	-1.024	0.308
	EN-4	-0.659	0.511	-1.034	0.303	**-2.025**	**0.045**	-1.228	0.221	-1.077	0.283
	EN-5	-0.803	0.423	-1.028	0.305	-1.120	0.264	-1.356	0.177	-1.119	0.265
180	EN-1	-0.577	0.565	-1.022	0.308	-1.060	0.291	-0.430	0.668	-1.136	0.258
	EN-2	-0.425	0.672	-1.068	0.287	-1.145	0.254	-0.448	0.655	-1.144	0.254
	EN-3	-0.546	0.586	-1.048	0.296	-1.073	0.285	-0.545	0.586	-1.089	0.278
	EN-4	-0.471	0.638	-1.089	0.278	-1.167	0.245	-0.725	0.469	-1.088	0.278
	EN-5	0.628	0.531	-1.287	0.200	-1.222	0.223	-0.749	0.455	-1.184	0.238
240	EN-1	-0.859	0.392	-1.096	0.275	-1.178	0.240	-0.157	0.875	-0.524	0.601
	EN-2	-0.889	0.375	0.451	0.653	-1.536	0.126	-0.231	0.818	-1.334	0.184
	EN-3	-0.912	0.363	-1.082	0.281	-1.323	0.188	-0.348	0.729	-1.360	0.176
	EN-4	-0.916	0.361	0.546	0.586	-1.502	0.135	-0.488	0.626	-1.306	0.194
	EN-5	-0.778	0.438	0.055	0.956	-1.377	0.171	-0.126	0.900	-1.392	0.166

Notes: Each Elastic Net model is compared only with AR(1). Reported statistics include the Harvey–Leybourne–Newbold correction and use two-sided p-values. Positive statistics favour the Elastic Net model and negative statistics favour AR(1). Bold values indicate significance at the 5% level.

The DM results give little evidence that the Elastic Net models improve on AR(1). For the main 180-month window, none of the five Elastic Net models is significantly different from AR(1) at any horizon. This includes EN-5 at one month. Although EN-5 has a lower MSFE than AR(1), its DM statistic is 0.628 with a p-value of 0.531. The numerical improvement is therefore not statistically significant.

The same result generally holds for the other windows. With the 120-month window, EN-3 has a lower MSFE than AR(1) at six months, but the DM test is not significant ($DM = 0.505$, $p = 0.614$). At the same horizon, EN-4 has a significantly negative DM statistic of -2.025

($p = 0.045$). Because a negative statistic favours AR(1), EN-4 performs significantly worse than the benchmark.

With the 240-month window, the Elastic Net models that slightly outperform AR(1) at three months are also not significantly different from the benchmark. EN-2 has $DM = 0.451$ ($p = 0.653$), EN-4 has $DM = 0.546$ ($p = 0.586$), and EN-5 has $DM = 0.055$ ($p = 0.956$). None provides significant evidence of better predictive accuracy.

AR(1) has the lowest forecast error in most horizon–window combinations. The few numerical gains from the Elastic Net models are not statistically significant. This confirms that AR(1) is difficult to outperform in conditional-mean forecasting.

The lack of a significant improvement over AR(1) does not mean that the financial variables have no predictive information. They may still add forecasting value within the Elastic Net models even when the full model does not outperform AR(1). Therefore, the next question is whether the S&P 500 return, Treasury information, and their interaction add information beyond smaller Elastic Net models. The Clark–West test is used for this comparison.

4.4 Incremental Predictive Content of Financial Information: Clark–West Tests

The Clark–West test is used to examine the incremental predictive content of the financial variables. The test compares valid nested Elastic Net models. It shows whether adding financial information improves the predictive performance of a smaller model. The analysis uses the main 180-month rolling window. AR(1) is not included in these comparisons. The Clark–West test uses an MSFE adjustment to account for the additional estimation noise in the larger model.

Table 4: MSFE-Adjusted Clark–West Tests for Nested Elastic Net Models

		1 Month		3 Months		6 Months		12 Months		24 Months	
Smaller	**Larger**	**CW**	p	**CW**	p	**CW**	p	**CW**	p	**CW**	p
EN-1	EN-2	0.408	0.342	1.022	0.153	0.962	0.168	0.066	0.474	-0.746	0.772
EN-1	EN-3	0.792	0.214	-1.750	0.960	-0.929	0.824	-0.528	0.701	**1.431**	**0.076**
EN-1	EN-4	0.843	0.200	1.019	0.154	-0.521	0.699	-0.630	0.736	0.833	0.202
EN-1	EN-5	1.080	0.140	1.015	0.155	-1.081	0.860	-0.723	0.765	0.922	0.178
EN-2	EN-4	0.496	0.310	0.676	0.250	-1.008	0.843	-0.974	0.835	**1.528**	**0.063**
EN-2	EN-5	0.835	0.202	0.980	0.164	-1.000	0.841	-1.073	0.858	**1.490**	**0.068**
EN-3	EN-4	1.146	0.126	1.048	0.147	0.861	0.195	-0.630	0.736	-1.207	0.886
EN-3	EN-5	1.276	0.101	1.040	0.149	-0.624	0.734	-0.828	0.796	-1.358	0.913
EN-4	EN-5	1.271	0.102	0.997	0.159	-1.162	0.877	-1.071	0.858	0.326	0.372

Notes: CW denotes the Clark–West MSFE-adjusted statistic and p is the corresponding one-sided p-value. Positive statistics favour the larger nested model. Bold values indicate significance at the 10% level. No comparison is significant at the 5% level.

Table 4 reports the Clark–West results for all valid nested Elastic Net comparisons. The results show limited evidence of incremental predictive content from the financial variables. None of the comparisons is significant at the 10

At the 24-month horizon, adding Treasury information to EN-1 produces a Clark–West statistic of 1.431 and a one-sided p-value of 0.076. Adding Treasury information to EN-2 to form EN-4 gives $CW = 1.528$ and $p = 0.063$. The comparison between EN-2 and EN-5 gives $CW = 1.490$ and $p = 0.068$. These three comparisons are significant at the 10

The main research question focuses on the interaction between S&P 500 returns and Treasury movements. This is examined directly by comparing EN-4 with EN-5. EN-4 contains the Restricted FRED-MD predictors, the S&P 500 return, and Treasury information. EN-5 contains the same predictors and additionally includes the S&P 500–Treasury interaction. Table 5 reports this comparison in more detail.

Table 5: Clark–West Test for the S&P 500–Treasury Interaction

	1 Month	**3 Months**	**6 Months**	**12 Months**	**24 Months**
MSFE without interaction	2.7610	3.0292	2.6611	2.0788	3.5292
MSFE with interaction	2.4567	2.7412	2.8623	2.0796	3.6379
MSFE-adjusted difference	0.3876	0.3178	-0.1505	-0.0007	0.0371
CW statistic	1.2708	0.9975	-1.1619	-1.0715	0.3259
One-sided p-value	0.1019	0.1593	0.8774	0.8580	0.3723

Notes: EN-5 differs from EN-4 through the addition of the S&P 500–Treasury interaction. Positive Clark–West statistics favour EN-5. Reported p-values are one-sided.

Table 5 shows that the largest numerical improvement from the interaction occurs at the one-month horizon. The MSFE decreases from 2.7610 for EN-4 to 2.4567 for EN-5. The MSFE-adjusted difference is also positive at 0.3876. The Clark–West statistic is 1.2708 with a one-sided p-value of 0.1019. The result is therefore close to the 10

At the three-month horizon, the interaction also reduces the MSFE. It falls from 3.0292 to 2.7412. However, the Clark–West statistic is 0.9975 with a p-value of 0.1593. The improvement is therefore not statistically significant.

At the 6- and 12-month horizons, the Clark–West statistics are negative. The corresponding p-values are 0.8774 and 0.8580. At the 24-month horizon, the Clark–West statistic is positive but small at 0.3259, with a p-value of 0.3723. These results provide no statistically significant evidence that adding the interaction improves predictive performance at these horizons.

Overall, the Clark–West results provide weak evidence that Treasury-related information adds predictive content at the 24-month horizon. However, the direct EN-4 versus EN-5 comparison shows that the S&P 500–Treasury interaction does not provide statistically significant incremental predictive information at any forecast horizon.

4.5 Main Findings from the Point Forecasts

The point-forecast results show a clear pattern. The AR(1) model performs well across most forecast horizons and rolling windows. The Elastic Net models perform better in a few cases. However, the Diebold–Mariano tests show that these improvements are not statistically significant.

The Clark–West tests also show limited evidence of improvement. Treasury information adds some predictive information at the 24-month horizon. However, this evidence is weak. The S&P 500–Treasury interaction does not add significant predictive information at any forecast horizon.

Overall, the financial variables do not consistently improve the point forecasts of industrial production growth. However, point forecasts focus on average future growth. Financial information may be more useful when industrial production growth is unusually low or high. The next section examines this possibility using quantile forecasting.

5 Additional Analysis: Quantile Forecasting

Quantile forecasting is used as an additional analysis to examine whether financial information has different predictive value across the growth distribution. Seven quantiles, $\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$, are evaluated at the 1-, 3-, 6-, 12-, and 24-month horizons using the main 180-month rolling window.

- **QAR(1)**
- **Q-EN-1:** Restricted FRED-MD predictors
- **Q-EN-2:** Restricted FRED-MD predictors + S&P 500 return
- **Q-EN-3:** Restricted FRED-MD predictors + daily 10-year Treasury log-change
- **Q-EN-4:** Restricted FRED-MD predictors + daily S&P 500 return + daily Treasury log-change
- **Q-EN-5:** Restricted FRED-MD predictors + daily S&P 500 return + daily Treasury log-change + daily S&P 500 × Treasury interaction

5.1 Quantile Forecasting Approach

The Quantile Autoregressive model, QAR(1), is used as the benchmark (Koenker and Xiao, 2006):

$$Q_\tau(y_{t+h} \mid y_t) = c_{h,\tau} + \phi_{h,\tau} y_t. \tag{18}$$

Quantile Elastic Net (Q-EN) combines quantile regression with regularization (Su and Wang, 2021). For forecast horizon h and quantile τ, the estimator is

$$(\hat{\beta}_{0,h,\tau}, \hat{\boldsymbol{\beta}}_{h,\tau}) = \arg\min_{\beta_0,\boldsymbol{\beta}} \left\{ \sum_{t \in \mathcal{T}_h} \rho_\tau(y_{t+h} - \beta_0 - \mathbf{x}_t'\boldsymbol{\beta}) + N_h \lambda_1 \|\boldsymbol{\beta}\|_1 + \frac{N_h \lambda_2}{2} \|\boldsymbol{\beta}\|_2^2 \right\}. \tag{19}$$

Forecast accuracy is measured using average pinball loss,

$$\overline{L}_{h,\tau} = \frac{1}{N_{h,\tau}} \sum_{t=1}^{N_{h,\tau}} \rho_\tau \left(y_{t+h} - \hat{Q}_{\tau,t+h|t} \right), \tag{20}$$

where a lower loss means a more accurate quantile forecast. The model specifications use the same information sets as the point forecasts.

5.2 Quantile Forecast Results

Table 6 shows the best-performing model at each quantile and horizon. Relative loss is measured against QAR(1), so a value below one means that the model has a lower loss than the benchmark.

The main difference from the point forecasts appears in the lower tail. At $\tau = 0.05$, Q-EN models have lower loss than QAR(1) at the 1-, 6-, and 12-month horizons. The largest numerical

Table 6: Best Quantile Forecasts: 180-Month Rolling Window

	1 Month			3 Months			6 Months			12 Months			24 Months		
τ	**Model**	**Loss**	**Rel.**	**Model**	**Loss**	**Rel.**	**Model**	**Loss**	**Rel.**	**Model**	**Loss**	**Rel.**	**Model**	**Loss**	**Rel.**
0.05	Q-EN-2	0.181	0.918	QAR(1)	0.198	1.000	Q-EN-3	0.172	0.737	Q-EN-5	0.193	0.896	QAR(1)	0.237	1.000
0.10	QAR(1)	0.218	1.000	Q-EN-3	0.245	0.996	Q-EN-3	0.236	0.986	QAR(1)	0.231	1.000	Q-EN-2	0.250	0.967
0.25	QAR(1)	0.292	1.000	QAR(1)	0.316	1.000	Q-EN-3	0.300	0.996	QAR(1)	0.309	1.000	QAR(1)	0.326	1.000
0.50	QAR(1)	0.344	1.000	QAR(1)	0.352	1.000	QAR(1)	0.344	1.000	QAR(1)	0.360	1.000	QAR(1)	0.360	1.000
0.75	QAR(1)	0.269	1.000	QAR(1)	0.274	1.000	QAR(1)	0.281	1.000	QAR(1)	0.298	1.000	QAR(1)	0.292	1.000
0.90	QAR(1)	0.177	1.000	QAR(1)	0.174	1.000	QAR(1)	0.187	1.000	QAR(1)	0.197	1.000	QAR(1)	0.194	1.000
0.95	QAR(1)	0.129	1.000	QAR(1)	0.150	1.000	Q-EN-3	0.130	0.961	Q-EN-1	0.138	0.975	QAR(1)	0.143	1.000

Note: Loss denotes average pinball loss. Relative loss is measured against QAR(1); values below one indicate lower forecast loss.

gain occurs at six months, where Q-EN-3, which includes Treasury information, has a relative loss of 0.737. At 12 months, Q-EN-5 has the lowest loss, with a relative loss of 0.896. This gives some evidence that the S&P 500–Treasury interaction may be useful when forecasting unusually weak industrial production growth.

This pattern does not appear across the full distribution. QAR(1) is the best model at every horizon for $\tau = 0.50$, 0.75, and 0.90. The larger information set therefore appears more useful for selected downside forecasts than for typical or stronger growth outcomes.

5.3 Predictive Accuracy Tests

The Diebold–Mariano tests compare each Q-EN model with QAR(1) at the same quantile and forecast horizon. The full results are reported in Appendix C. Table 7 shows the main results.

Table 7: Selected Diebold–Mariano Test Results

Quantile	**Horizon**	**Model**	**DM**	p**-value**
0.05	6 Months	Q-EN-3	1.099	0.274
0.05	12 Months	Q-EN-5	0.816	0.416
0.75	1 Month	Q-EN-1	-1.992	**0.048**
0.75	1 Month	Q-EN-3	-1.884	0.061
0.75	1 Month	Q-EN-4	-1.832	0.069
0.75	1 Month	Q-EN-5	-1.714	0.089

Notes: Reported statistics are HLN-adjusted DM statistics with two-sided p-values. Positive values favour Q-EN and negative values favour QAR(1). The full set of results is reported in Appendix C.

The DM tests give no significant evidence that the Q-EN models outperform QAR(1). The positive results in the 5% lower tail are not statistically significant, including Q-EN-3 at six months ($p = 0.274$) and Q-EN-5 at 12 months ($p = 0.416$). The strongest significant result occurs at the one-month 75% quantile, where the negative statistics favour QAR(1). Q-EN-1 is significantly worse at the 5% level ($p = 0.048$), while several other Q-EN models are significant at the 10% level. Overall, the tests do not show a systematic predictive advantage for the Q-EN models.

5.4 Comparison with Point Forecasts

The quantile analysis gives a complementary view of the point-forecast results. In the point forecasts, AR(1) has the lowest MSFE in most horizon–window combinations, and the occasional numerical gains from the Elastic Net models are not statistically significant. Thus, there is little evidence that adding the S&P 500 return, Treasury information, or their interaction consistently improves conditional-mean forecasts.

The quantile results suggest that the value of financial information may depend on the part of the industrial production growth distribution being forecast. At the 5% quantile, Q-EN models have lower pinball loss than QAR(1) at selected horizons. The largest numerical gain occurs at six months, where Q-EN-3 has a relative loss of 0.737. At 12 months, Q-EN-5 has the lowest loss, with a relative loss of 0.896. In contrast, QAR(1) generally performs better near the centre and in the upper part of the distribution.

These results suggest that financial information may be more useful for forecasting unusually weak industrial production growth than for outcomes near the centre or upper part of the distribution. However, the Diebold–Mariano tests do not show significant improvements over QAR(1) for the lower-tail results. The evidence is therefore suggestive rather than conclusive and does not change the main point-forecast result that AR(1) is difficult to outperform.

6 Conclusion

The point-forecast results show that AR(1) is difficult to outperform. It has the lowest MSFE in most horizon–window combinations, and the occasional numerical gains from the Elastic Net models are not statistically significant according to the Diebold–Mariano tests. The Clark–West tests give weak evidence that Treasury-related information adds predictive content at the 24-month horizon, but only at the 10% significance level. The S&P 500–Treasury interaction does not add statistically significant predictive information at any of the five horizons. Therefore, the main research question is not supported by the point-forecast results.

The quantile analysis gives a different but limited result. Financial information produces lower forecast loss in some lower-tail cases. Treasury information performs well at the 6-month 5% quantile, while the interaction model performs best at the 12-month 5% quantile. However, the Diebold–Mariano tests do not show that these improvements are statistically significant. Financial information may therefore be more useful for forecasting unusually weak industrial production outcomes, but the evidence is only suggestive.

A larger information set does not always produce better out-of-sample forecasts. The simple autoregressive benchmark remains competitive even with a large macroeconomic and financial predictor set. The results also show that the value of financial information can depend on the forecast horizon and on the part of the industrial production growth distribution being forecast.

6.1 Limitations

The results have several limitations. First, the Elastic Net models contain many Restricted FRED-MD predictors together with the financial variables. Although regularization controls

model complexity, the large predictor set can make it difficult to isolate the contribution of individual financial variables. Second, the sample includes the COVID-19 period, when industrial production had unusually large movements that may affect forecast errors and model comparisons. Third, the analysis focuses on the S&P 500 and the 10-year Treasury yield and does not include all types of financial information that may help forecast industrial production.

6.2 Future Research

Future research can extend the analysis in several ways. An expanding window can be compared with the rolling-window approach to test whether a growing historical sample changes forecast performance. A fully real-time analysis using historical data vintages and publication lags would provide a stricter test of the forecasting results. The financial information set can also be expanded to include other market variables.

The lower-tail results also provide a direction for future research. Since the quantile forecasts show some numerical gains during weak industrial production outcomes, future work can study downside forecasting in more detail and test whether these patterns remain in other samples or models. Other regularization and machine-learning methods can also be compared with AR(1) and Elastic Net to test whether more flexible models can extract additional predictive information.

References

Board of Governors of the Federal Reserve System (US) (2026). Market yield on u.s. treasury securities at 10-year constant maturity, quoted on an investment basis [dgs10]. Federal Reserve Economic Data (FRED), Federal Reserve Bank of St. Louis. Retrieved from FRED.

Chen, N.-F., Roll, R., and Ross, S. A. (1986). Economic forces and the stock market. *Journal of Business*, 59(3):383–403.

Clark, T. E. and West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1):291–311.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.

Fama, E. F. (1990). Stock returns, expected returns, and real activity. *Journal of Finance*, 45(4):1089–1108.

Harvey, D., Leybourne, S., and Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291.

Hyndman, R. J. and Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*. OTexts, Melbourne, Australia, 3 edition.

Koenker, R. and Xiao, Z. (2006). Quantile autoregression. *Journal of the American Statistical Association*, 101(475):980–990.

McCracken, M. W. and Ng, S. (2016). Fred-md: A monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4):574–589.

Ratanapakorn, O. and Sharma, S. C. (2007). Dynamic analysis between the us stock returns and the macroeconomic variables. *Applied Financial Economics*, 17(5):369–377.

Su, M. and Wang, W. (2021). Elastic net penalized quantile regression model. *Journal of Computational and Applied Mathematics*, 392:113462.

Theil, H. (1966). *Applied Economic Forecasting*. North-Holland Publishing Company, Amsterdam.

West, K. D. (2006). Forecast evaluation. In Elliott, G., Granger, C. W. J., and Timmermann, A., editors, *Handbook of Economic Forecasting*, volume 1, pages 99–134. Elsevier.

Yahoo Finance (2026). S&p 500 historical data. https://finance.yahoo.com/quote/%5EGSPC/history/. Accessed: 2026.

Zou, H. and Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301–320.

A FRED-MD Variables by Economic Sector

The FRED-MD variables are classified into eight economic groups following McCracken and Ng (2016). Table 8 gives the FRED mnemonic and a short description of each variable.

Table 8: FRED-MD Variables by Economic Sector

Variable	Description	Variable	Description	Variable	Description	Variable	Description
Group 1: Output and Income							
RPI	Real Personal Income	W875RX1	Real Personal Income Excluding Transfer Receipts	INDPRO	Industrial Production Index	IPFPNSS	Industrial Production: Final Products and Nonindustrial Supplies
IPFINAL	Industrial Production: Final Products	IPCONGD	Industrial Production: Consumer Goods	IPDCONGD	Industrial Production: Durable Consumer Goods	IPNCONGD	Industrial Production: Nondurable Consumer Goods
IPBUSEQ	Industrial Production: Business Equipment	IPMAT	Industrial Production: Materials	IPDMAT	Industrial Production: Durable Materials	IPNMAT	Industrial Production: Nondurable Materials
IPMANSICS	Industrial Production: Manufacturing	IPB51222s	Industrial Production: Residential Utilities	IPFUELS	Industrial Production: Fuels	NAPMPI	ISM Manufacturing: Production Index
CUMFNS	Capacity Utilization: Manufacturing						
Group 2: Labor Market							
HWI	Help-Wanted Index for the United States	HWIURATIO	Ratio of Help Wanted to Number of Unemployed	CLF16OV	Civilian Labor Force	CE16OV	Civilian Employment
UNRATE	Civilian Unemployment Rate	UEMPMEAN	Average Duration of Unemployment	UEMPLT5	Unemployed for Less Than 5 Weeks	UEMP5TO14	Unemployed for 5–14 Weeks
UEMP15OV	Unemployed for 15 Weeks and Over	UEMP15T26	Unemployed for 15–26 Weeks	UEMP27OV	Unemployed for 27 Weeks and Over	CLAIMSx	Initial Unemployment Insurance Claims
PAYEMS	Total Nonfarm Employment	USGOOD	Employment: Goods-Producing Industries	CES1021000001	Employment: Mining	USCONS	Employment: Construction
MANEMP	Employment: Manufacturing	DMANEMP	Employment: Durable Goods	NDMANEMP	Employment: Nondurable Goods	SRVPRD	Employment: Service-Providing Industries
USTPU	Employment: Trade, Transportation and Utilities	USWTRADE	Employment: Wholesale Trade	USTRADE	Employment: Retail Trade	USFIRE	Employment: Financial Activities
USGOVT	Government Employment	CES0600000007	Average Weekly Hours: Goods-Producing	AWOTMAN	Average Weekly Overtime Hours: Manufacturing	AWHMAN	Average Weekly Hours: Manufacturing
NAPMEI	ISM Manufacturing: Employment Index	CES0600000008	Average Hourly Earnings: Goods-Producing	CES2000000008	Average Hourly Earnings: Construction	CES3000000008	Average Hourly Earnings: Manufacturing
Group 3: Consumption and Housing							
HOUST	Housing Starts: Total	HOUSTNE	Housing Starts: Northeast	HOUSTMW	Housing Starts: Midwest	HOUSTS	Housing Starts: South
HOUSTW	Housing Starts: West	PERMIT	New Private Housing Permits: Total	PERMITNE	Housing Permits: Northeast	PERMITMW	Housing Permits: Midwest
PERMITS	Housing Permits: South	PERMITW	Housing Permits: West				
Group 4: Orders and Inventories							
DPCERA3M086SBEA	Real Personal Consumption Expenditures	CMRMTSPLx	Real Manufacturing and Trade Industries Sales	RETAILx	Retail and Food Services Sales	NAPM	ISM Manufacturing: PMI Composite Index
NAPMNOI	ISM Manufacturing: New Orders Index	NAPMSDI	ISM Manufacturing: Supplier Deliveries Index	NAPMII	ISM Manufacturing: Inventories Index	ACOGNO	New Orders for Consumer Goods
AMDMNOx	New Orders for Durable Goods	ANDENOx	New Orders for Nondefense Capital Goods	AMDMUOx	Unfilled Orders for Durable Goods	BUSINVx	Total Business Inventories
ISRATIOx	Business Inventories-to-Sales Ratio	UMCSENTx	Consumer Sentiment Index				

Variable	Description	Variable	Description	Variable	Description	Variable	Description
Group 5: Money and Credit							
M1SL	M1 Money Stock	M2SL	M2 Money Stock	M2REAL	Real M2 Money Stock	AMBSL	St. Louis Adjusted Monetary Base
TOTRESNS	Total Reserves of Depository Institutions	NONBORRES	Nonborrowed Reserves of Depository Institutions	BUSLOANS	Commercial and Industrial Loans	REALLN	Real Estate Loans at Commercial Banks
NONREVSL	Total Nonrevolving Consumer Credit	CONSPI	Nonrevolving Consumer Credit to Personal Income	MZMSL	MZM Money Stock	DTCOLNVHFNM	Consumer Motor Vehicle Loans Outstanding
DTCTHFNM	Total Consumer Loans and Leases Outstanding	INVEST	Securities in Bank Credit at Commercial Banks				
Group 6: Interest Rates and Exchange Rates							
FEDFUNDS	Effective Federal Funds Rate	CP3Mx	3-Month AA Financial Commercial Paper Rate	TB3MS	3-Month Treasury Bill Rate	TB6MS	6-Month Treasury Bill Rate
GS1	1-Year Treasury Rate	GS5	5-Year Treasury Rate	GS10	10-Year Treasury Rate	AAA	Moody's Seasoned Aaa Corporate Bond Yield
BAA	Moody's Seasoned Baa Corporate Bond Yield	COMPAPFFx	Commercial Paper Rate Minus Federal Funds Rate	TB3SMFFM	3-Month Treasury Rate Minus Federal Funds Rate	TB6SMFFM	6-Month Treasury Rate Minus Federal Funds Rate
T1YFFM	1-Year Treasury Rate Minus Federal Funds Rate	T5YFFM	5-Year Treasury Rate Minus Federal Funds Rate	T10YFFM	10-Year Treasury Rate Minus Federal Funds Rate	AAAFFM	Aaa Corporate Bond Yield Minus Federal Funds Rate
BAAFFM	Baa Corporate Bond Yield Minus Federal Funds Rate	TWEXMMTH	Trade-Weighted U.S. Dollar Index	EXSZUSx	Switzerland/U.S. Exchange Rate	EXJPUSx	Japan/U.S. Exchange Rate
EXUSUKx	U.S./U.K. Exchange Rate	EXCAUSx	Canada/U.S. Exchange Rate				
Group 7: Prices							
PPIFGS	Producer Price Index: Finished Goods	PPIFCG	Producer Price Index: Finished Consumer Goods	PPIITM	Producer Price Index: Intermediate Materials	PPICRM	Producer Price Index: Crude Materials
OILPRICEx	Crude Oil Price	PPICMM	Producer Price Index: Metals and Metal Products	NAPMPRI	ISM Manufacturing: Prices Index	CPIAUCSL	Consumer Price Index: All Items
CPIAPPSL	Consumer Price Index: Apparel	CPITRNSL	Consumer Price Index: Transportation	CPIMEDSL	Consumer Price Index: Medical Care	CUSR0000SAC	Consumer Price Index: Commodities
CUUR0000SAD	Consumer Price Index: Durables	CUSR0000SAS	Consumer Price Index: Services	CPIULFSL	Consumer Price Index: All Items Less Food	CUUR0000SA0L2	Consumer Price Index: All Items Less Shelter
CUSR0000SA0L5	Consumer Price Index: All Items Less Medical Care	PCEPI	Personal Consumption Expenditures Price Index	DDURRG3M086SBEA	PCE Price Index: Durable Goods	DNDGRG3M086SBEA	PCE Price Index: Nondurable Goods
DSERRG3M086SBEA	PCE Price Index: Services						
Group 8: Stock Market							
S&P 500	S&P's Common Stock Price Index: Composite	S&P: indust	S&P's Common Stock Price Index: Industrials	S&P div yield	S&P's Composite Common Stock: Dividend Yield	S&P PE ratio	S&P's Composite Common Stock: Price–Earnings Ratio

Note: The variables are grouped according to the FRED-MD classification of McCracken and Ng (2016). Variable names follow the FRED/FRED-MD mnemonics.

B FRED-MD Transformation Codes

The FRED-MD transformation codes are defined as follows:

Table 9: FRED-MD Transformation Codes

Code	Transformation	Definition
1	No transformation	x_t
2	First difference	$\Delta x_t = x_t - x_{t-1}$
3	Second difference	$\Delta^2 x_t = \Delta x_t - \Delta x_{t-1}$
4	Logarithm	$\log(x_t)$
5	First difference of logarithm	$\Delta \log(x_t) = \log(x_t) - \log(x_{t-1})$
6	Second difference of logarithm	$\Delta^2 \log(x_t) = \Delta \log(x_t) - \Delta \log(x_{t-1})$
7	Difference in percentage change	$\Delta(x_t/x_{t-1} - 1)$

Codes 1 and 4 keep the variable in levels or log levels. Codes 2 and 3 use first and second differences. Codes 5 and 6 apply the same differences after taking logarithms. Code 7 is the difference in the percentage growth rate.

C Diebold–Mariano Tests Against QAR(1)

Table 10: Diebold–Mariano Tests Against QAR(1): 180-Month Rolling Window

Quantile	Model	1 Month	3 Months	6 Months	12 Months	24 Months
0.05	Q-EN-1	0.554 (0.580)	-0.823 (0.412)	0.896 (0.372)	0.624 (0.534)	-1.079 (0.283)
	Q-EN-2	0.631 (0.529)	-0.823 (0.412)	1.058 (0.292)	0.621 (0.536)	-1.112 (0.268)
	Q-EN-3	0.513 (0.609)	-0.852 (0.396)	1.099 (0.274)	0.769 (0.443)	-1.134 (0.259)
	Q-EN-4	0.613 (0.541)	-0.852 (0.396)	0.959 (0.339)	0.772 (0.442)	-1.092 (0.277)
	Q-EN-5	0.580 (0.563)	-0.851 (0.396)	1.048 (0.296)	0.816 (0.416)	-1.081 (0.282)
0.10	Q-EN-1	-0.544 (0.587)	0.086 (0.932)	-0.090 (0.929)	-0.886 (0.377)	-1.155 (0.250)
	Q-EN-2	-0.463 (0.644)	0.086 (0.932)	-0.122 (0.903)	-0.886 (0.377)	0.475 (0.635)
	Q-EN-3	-0.563 (0.574)	0.102 (0.919)	0.346 (0.730)	-0.898 (0.371)	-1.099 (0.274)
	Q-EN-4	-0.571 (0.569)	0.102 (0.919)	0.246 (0.806)	-0.898 (0.371)	0.464 (0.644)
	Q-EN-5	-0.479 (0.632)	-0.458 (0.647)	-0.132 (0.895)	-0.833 (0.406)	0.485 (0.629)
0.25	Q-EN-1	-1.404 (0.162)	-1.095 (0.276)	0.039 (0.969)	-1.697 (0.092)	-1.263 (0.209)
	Q-EN-2	-1.259 (0.210)	-1.100 (0.273)	-0.196 (0.845)	-1.730 (0.086)	-1.439 (0.153)
	Q-EN-3	-1.140 (0.256)	-1.095 (0.275)	0.103 (0.918)	-1.481 (0.141)	-1.248 (0.214)
	Q-EN-4	-1.304 (0.194)	-1.101 (0.273)	-0.075 (0.941)	-1.710 (0.090)	-1.243 (0.216)
	Q-EN-5	-1.337 (0.183)	-0.919 (0.359)	-0.067 (0.947)	-1.455 (0.148)	-1.175 (0.242)
0.50	Q-EN-1	-0.634 (0.527)	-0.807 (0.421)	-0.272 (0.786)	-0.105 (0.917)	-1.357 (0.177)
	Q-EN-2	-0.813 (0.417)	-0.695 (0.488)	-0.261 (0.794)	-0.283 (0.778)	-1.421 (0.158)
	Q-EN-3	-1.273 (0.205)	-0.796 (0.427)	-0.038 (0.970)	-0.015 (0.988)	-1.048 (0.297)
	Q-EN-4	-0.472 (0.638)	-0.724 (0.470)	-0.394 (0.694)	-0.026 (0.979)	-1.106 (0.271)
	Q-EN-5	-0.781 (0.436)	-0.594 (0.554)	-0.156 (0.876)	-0.017 (0.986)	-1.068 (0.288)
0.75	Q-EN-1	-1.992 (0.048)	-1.423 (0.157)	-0.811 (0.419)	-1.245 (0.215)	-1.157 (0.250)
	Q-EN-2	-1.781 (0.077)	-1.433 (0.154)	-0.919 (0.360)	-1.213 (0.227)	-0.946 (0.346)
	Q-EN-3	-1.884 (0.061)	-1.078 (0.283)	-0.853 (0.395)	-1.177 (0.241)	-1.146 (0.254)
	Q-EN-4	-1.832 (0.069)	-1.087 (0.279)	-0.857 (0.393)	-1.215 (0.227)	-0.949 (0.344)
	Q-EN-5	-1.714 (0.089)	-1.150 (0.252)	-0.866 (0.388)	-1.155 (0.250)	-0.955 (0.341)
0.90	Q-EN-1	-1.481 (0.141)	-0.874 (0.384)	-0.557 (0.578)	-0.348 (0.728)	-0.807 (0.421)
	Q-EN-2	-1.642 (0.103)	-0.799 (0.426)	-1.018 (0.310)	-0.369 (0.712)	-0.780 (0.437)
	Q-EN-3	-0.649 (0.517)	-1.291 (0.199)	-0.637 (0.525)	-0.339 (0.735)	-0.802 (0.424)
	Q-EN-4	-1.597 (0.112)	-0.896 (0.372)	-0.953 (0.342)	-0.381 (0.704)	-0.778 (0.438)
	Q-EN-5	-1.123 (0.263)	-0.891 (0.375)	-0.954 (0.342)	-0.384 (0.702)	-0.779 (0.437)
0.95	Q-EN-1	-0.771 (0.442)	-0.091 (0.927)	1.075 (0.284)	0.604 (0.547)	-0.525 (0.600)
	Q-EN-2	-0.818 (0.415)	-0.068 (0.946)	0.837 (0.404)	0.596 (0.552)	-0.555 (0.580)
	Q-EN-3	-0.754 (0.452)	-0.123 (0.902)	1.272 (0.205)	0.596 (0.552)	-0.663 (0.509)
	Q-EN-4	-0.835 (0.405)	-0.119 (0.905)	1.054 (0.294)	0.583 (0.561)	-0.665 (0.507)
	Q-EN-5	-0.863 (0.390)	-0.115 (0.908)	1.054 (0.294)	0.581 (0.562)	-0.665 (0.507)

Note: Each cell reports the HLN-adjusted DM statistic with the two-sided p-value in parentheses. Positive values favour Q-EN and negative values favour QAR(1).